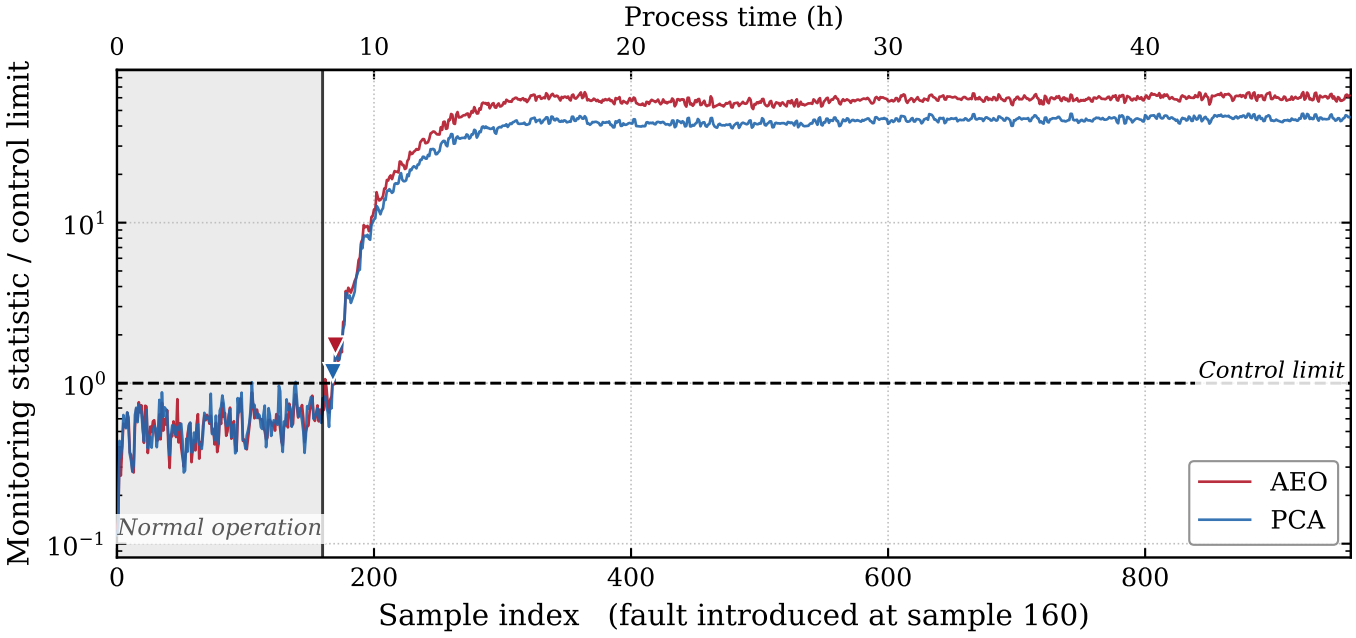
Fault Detection Performance for IDV(2)

Combined T^2 and SPE statistics — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	11.1 samples (33 min)	MDR 0.014	detected in 7/7 runs
PCA	delay	10.1 samples (30 min)	MDR 0.012	detected in 7/7 runs

PCA detects 1.0 samples (3 min) earlier